



sAlfer Lab

Joint lab on Safety and Security of AI

Computer Vision Technologies and Biometrics Fundamentals

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**M.Sc. Degree in Computer Engineering, CyberSecurity and
Artificial Intelligence**



University
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Department of Electrical
and Electronic
Engineering



Overview of the talk



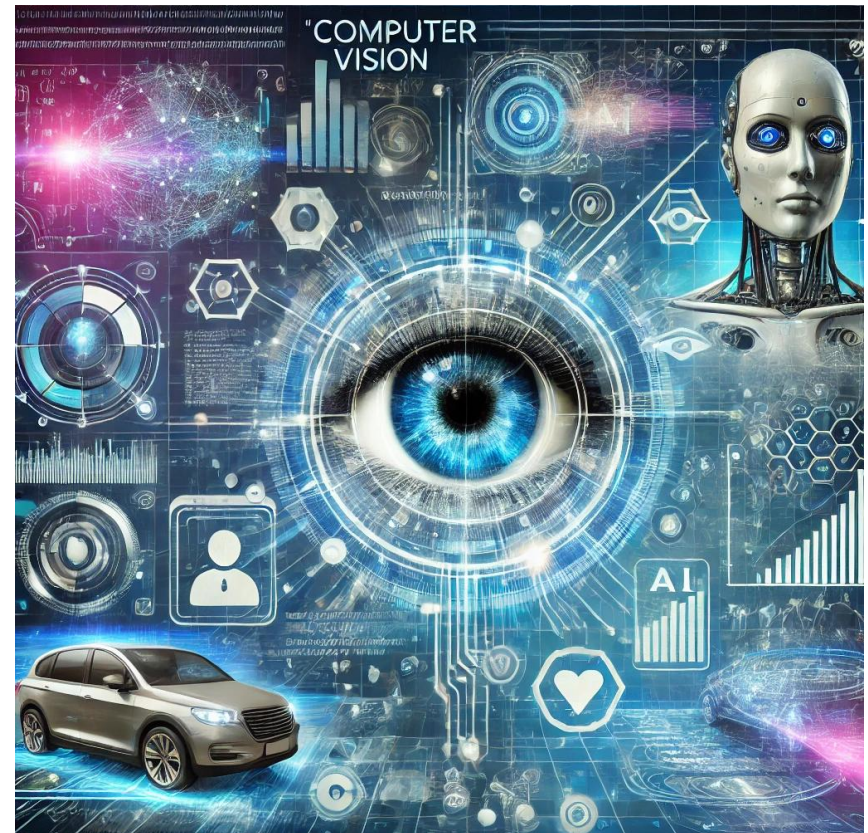
- **Definition**
- **Image formation and processing**
- **Random variables**

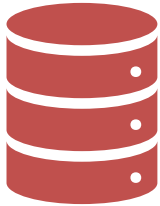
Basic definition



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- Replicating the human ability to
 - Scene representation
 - Object detection
 - Object interaction
 - Event classification & prediction





First level

Sensor: images and videos

- Low level representation
- Methods and algorithms for image processing
- Depth estimation

Object detection and tracking

- Classification of objects (static/dynamic)
- Interaction among objects when they are moving



Second level

Contextualization with respect to the application

- Objects become, for example, humans, cars, animals

Scene interpretation

- Movements of objects become actions (aggression/collision)
- Specific events

Why is it so difficult?



- **It is an inverse problem: to recover some unknowns given insufficient information to fully specify the solution.**
 - Physics-based and probabilistic models,
 - Machine learning from large sets of examples, to disambiguate between potential solutions
- **Computer vision aims to describe the world that we see in one or more images and to reconstruct its properties, such as shape, illumination, and color distributions.**

Industrial Applications



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Optical character recognition (OCR)

Machine inspection

Retail

Warehouse logistics

Medical imaging

Self-driving vehicles

3D model building (photogrammetry)

Match move

Motion capture

Surveillance

Biometrics

Consumer-level applications



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Stitching

Exposure bracketing

Morphing

3D modeling

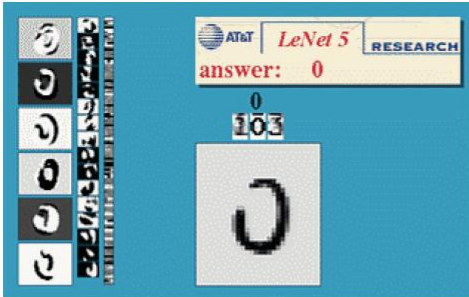
Video match move and stabilization

Photo-based walkthroughs

Face detection

Visual authentication

Examples



(a)



(b)



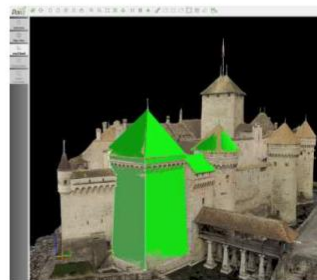
(c)



(d)



(e)



(f)

- (a) optical character recognition (OCR)
- (b) mechanical inspection
- (c) warehouse picking
- (d) medical imaging
- (e) self-driving cars
- (f) drone-based photogrammetry

Examples



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- (a) image stitching: merging different views
- (b) exposure bracketing: merging different exposures
- (c) morphing: blending between two photographs
- (d) smartphone augmented reality showing real-time depth occlusion effects



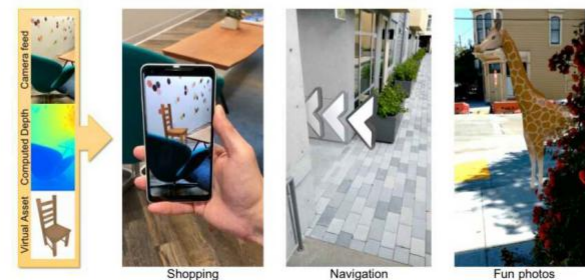
(a)



(b)



(c)



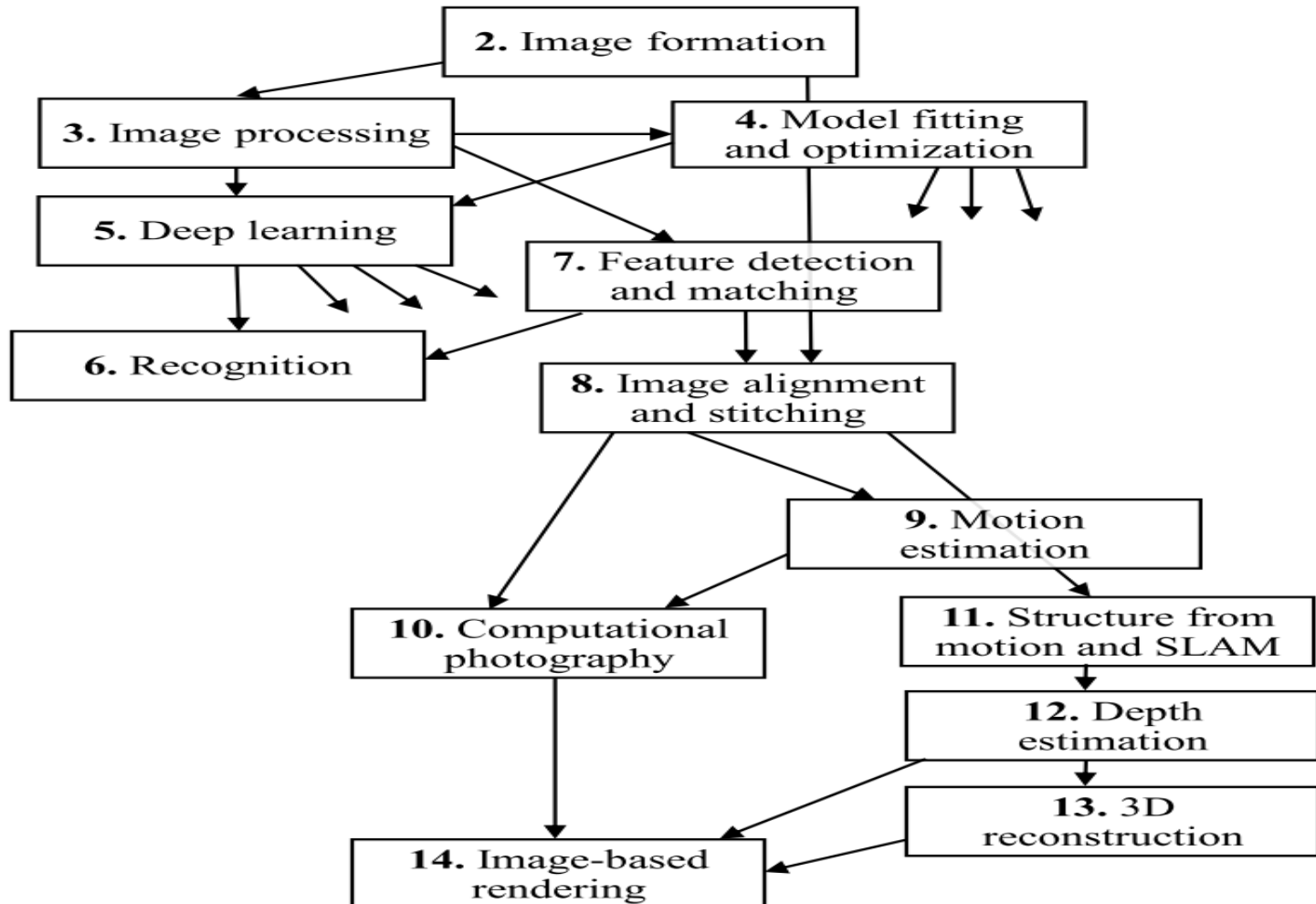
(d)

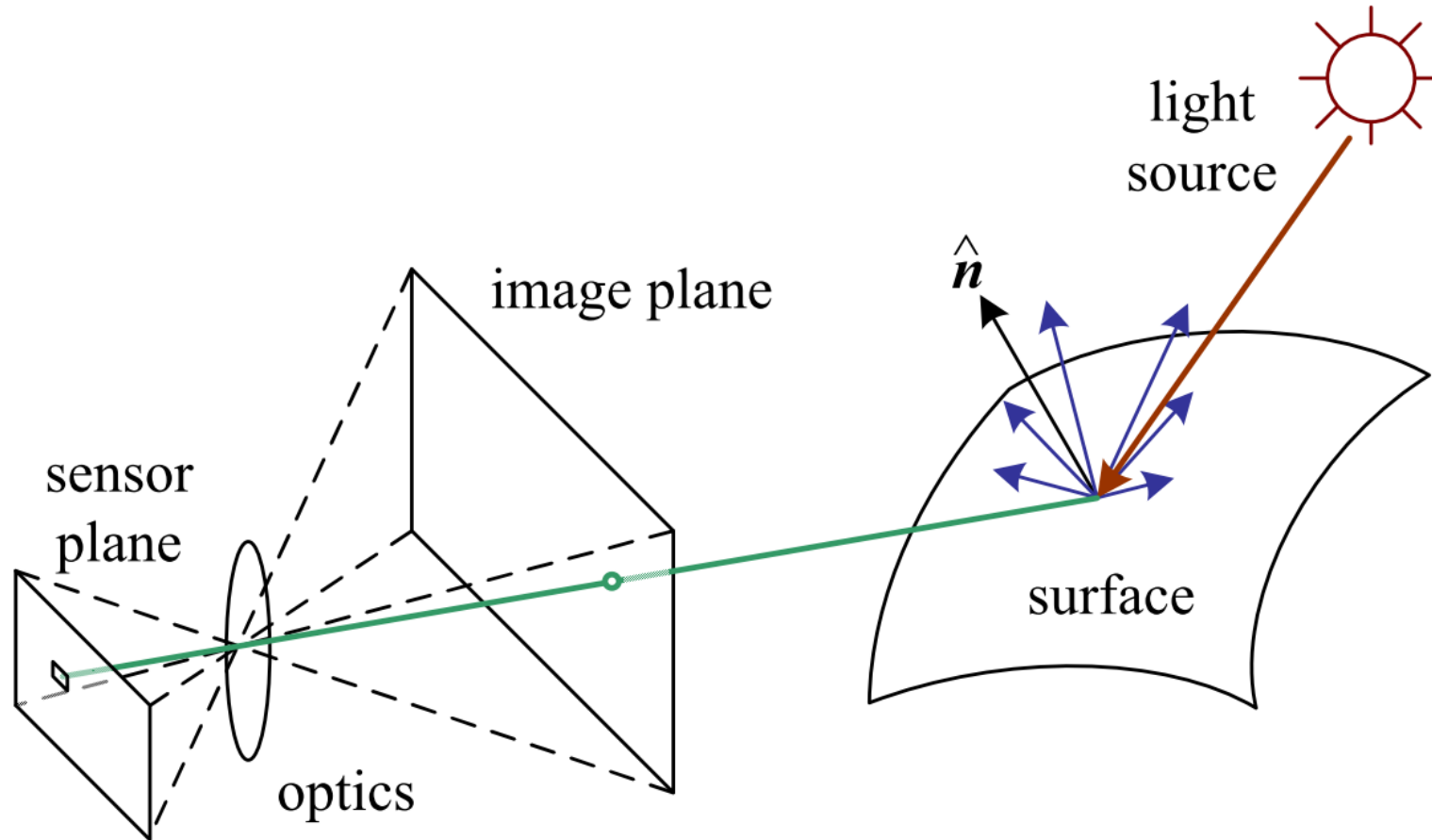
Topics in computer vision



2D (*what?*)

3D (*where?*)

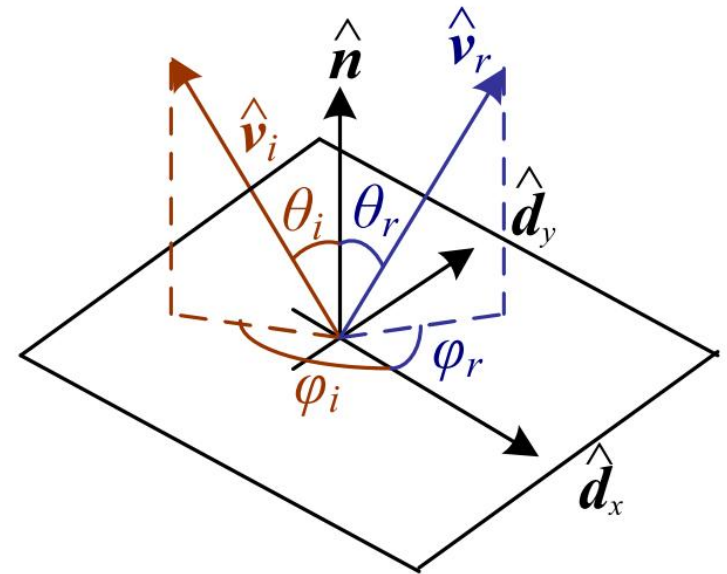




Light scattering



- **Bidirectional Reflectance Distribution**
 - how much of each wavelength arriving at an incident direction $\hat{v}_i$ is emitted in a reflected direction $\hat{v}_r$
- **Lambertian or diffuse reflection**
 - the smooth (non-shiny) variation of intensity with surface normal that is seen when observing a statue
- **Specular reflection**
 - The incident light ray direction $\hat{v}_i$ is reflected onto the specular direction around the surface normal $\hat{n}$.
- **Pong shading**
 - objects are generally illuminated not only by point light sources but also by a general diffuse illumination corresponding to inter-reflection (e.g., the walls in a room) or distant sources, such as the blue sky
 - Combines diffuse and specular reflections



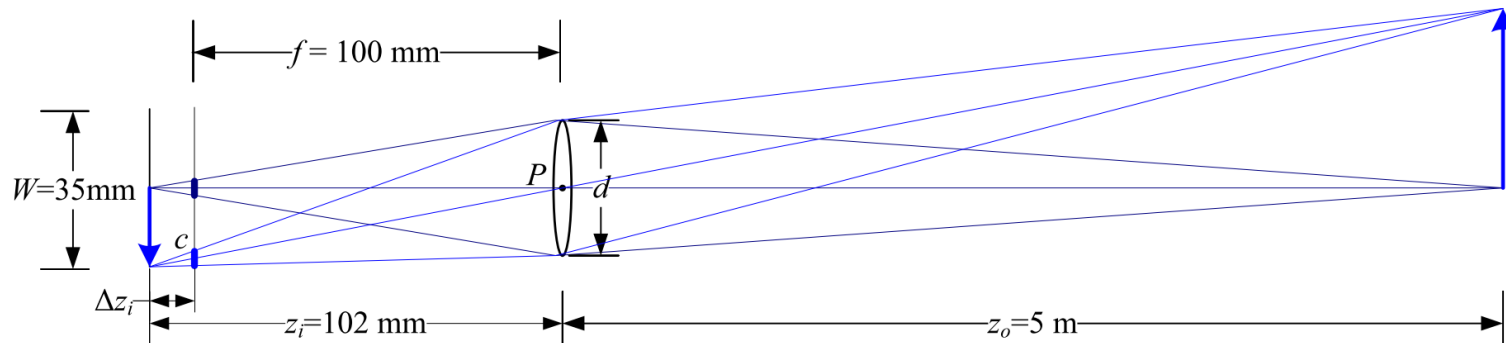
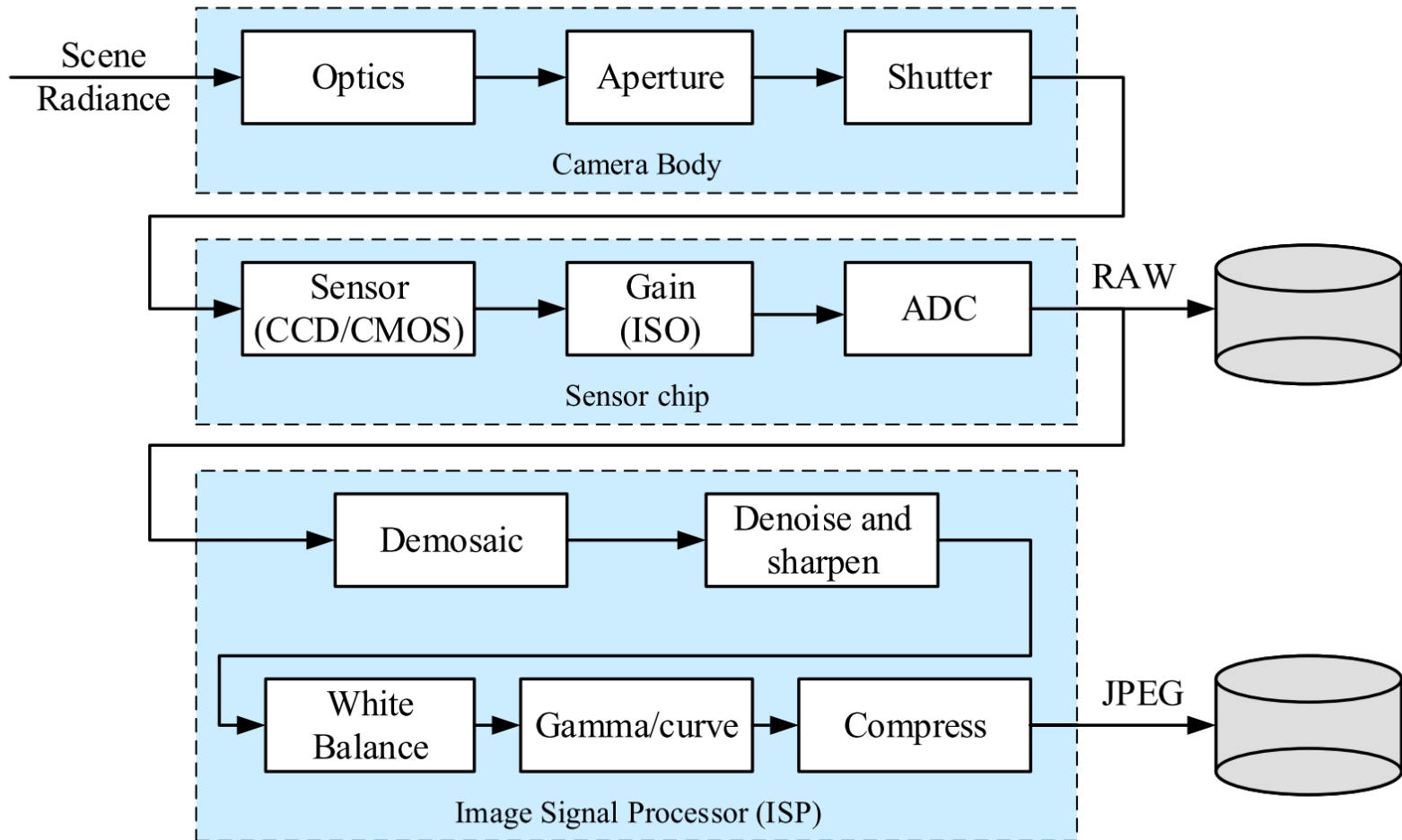


Figure 2.19 A thin lens of focal length f focuses the light from a plane at a distance z_o in front of the lens onto a plane at a distance z_i behind the lens, where $\frac{1}{z_o} + \frac{1}{z_i} = \frac{1}{f}$. If the focal plane (vertical gray line next to c) is moved forward, the images are no longer in focus and the circle of confusion c (small thick line segments) depends on the distance of the image plane motion Δz_i relative to the lens aperture diameter d . The field of view (f.o.v.) depends on the ratio between the sensor width W and the focal length f (or, more precisely, the focusing distance z_i , which is usually quite close to f).

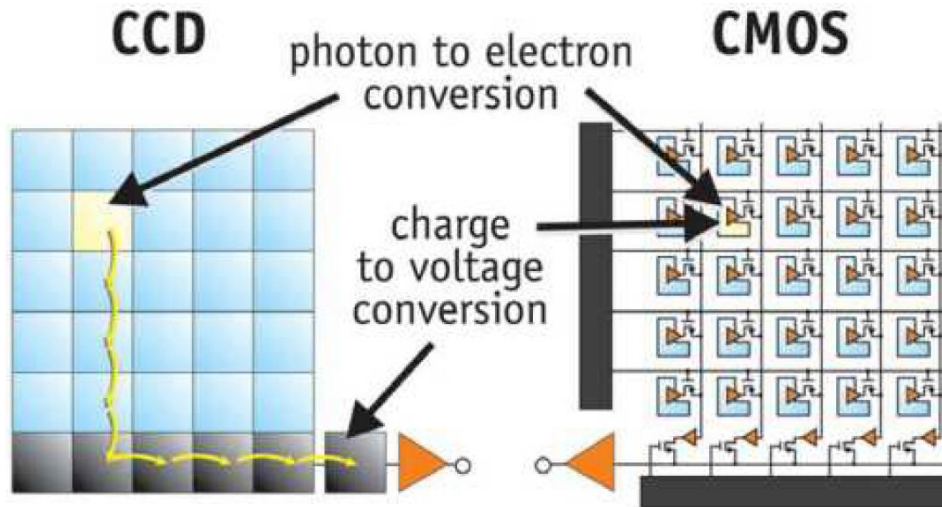
Image sensing



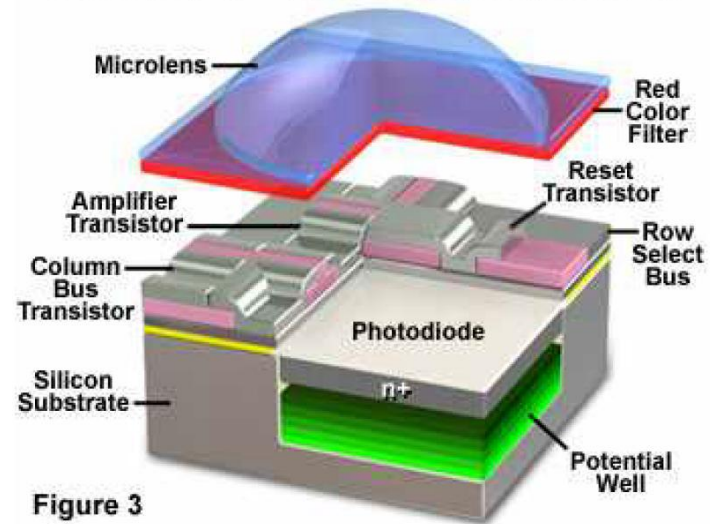
CCD/CMOS conversion



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Anatomy of the Active Pixel Sensor Photodiode



- CCDs move is photogenerated charge from pixel to pixel and convert it to voltage at the output node
- CMOS imagers convert charge to voltage inside each pixel by photodetector conductivity
- CCD sensors outperformed CMOS in quality-sensitive applications, such as digital SLRs, while CMOS was better for low-power applications, but today CMOS is used in most digital cameras

Image resolution



- Level of detail contained in an image: number of pixels that exist within that image
- Measuring resolution:
 - PPI Pixel-Per-Inch: the number of pixels contained within each inch of a digital image
 - DPI Dot-Per-Inch: the number of ink droplets a printer will produce per inch while printing an image
 - In many cases: PPI:DPI=1:1
- Inch: length-unit (2.54 cm)

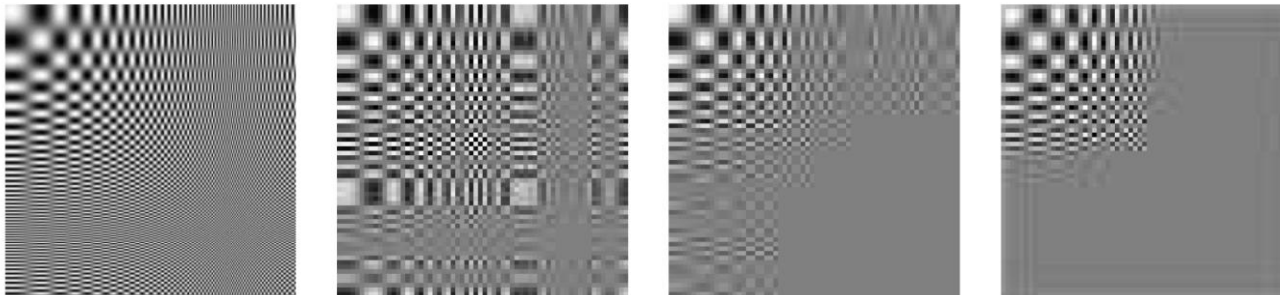
$$PPI = \frac{\sqrt{w^2 + h^2}}{d}$$

Length of the image diagonal

Design problems



- **Sensor noise**
- **Sampling and aliasing**
 - Shannon: $f_s \geq 2f_{max}$



- **Colors**
 - CIE/RGB
 - XYZ
 - Gamma compression

$$\begin{bmatrix} X \\ Y \\ Z \end{bmatrix} = \frac{1}{0.17697} \begin{bmatrix} 0.49 & 0.31 & 0.20 \\ 0.17697 & 0.81240 & 0.01063 \\ 0.00 & 0.01 & 0.99 \end{bmatrix} \begin{bmatrix} R \\ G \\ B \end{bmatrix}$$

$$Y' = Y^{\frac{1}{\gamma}}$$

YCbCr space



$$\begin{bmatrix} Y' \\ C_b \\ C_r \end{bmatrix} = \begin{bmatrix} 0.299 & 0.587 & 0.114 \\ -0.168736 & -0.331264 & 0.5 \\ 0.5 & -0.418688 & -0.081312 \end{bmatrix} \begin{bmatrix} R' \\ G' \\ B' \end{bmatrix} + \begin{bmatrix} 0 \\ 128 \\ 128 \end{bmatrix}$$

- **R'G'B'** values are the eight-bit gamma-compressed color components (i.e., the actual RGB values we obtain when we open up or display a JPEG image)
- For most applications, this formula is not that important, since image reading software will directly provide with the eight-bit gamma-compressed R'G'B' values

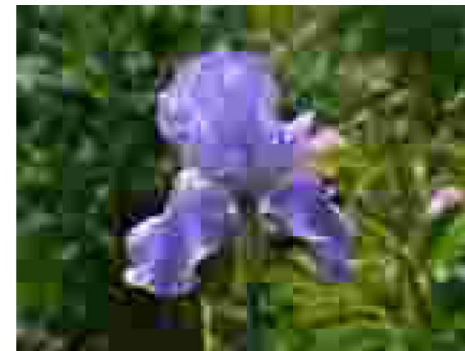




Image processing

Outline



- **Point operators**
- **Linear filtering**
- **Neighborhood operators**
- **Fourier transforms**
- **Wavelets**
- **Geometric transformations**

- Each output pixel's value depends on only the corresponding input pixel value
- Linear transformation of a pixel x :

$$g(\mathbf{x}) = af(\mathbf{x}) + b.$$

- $a > 0$, b are *contrast* and *brightness* parameters, respectively

- Histogram equalization

$$c(I) = \frac{1}{N} \sum_{i=0}^I h(i) = c(I-1) + \frac{1}{N} h(I)$$

- Goal: to find a mapping function $f(I)$ such that $h(I)$ is flat
- I and c are scaled in $[0, 255]$
- We can also compute a local histogram equalization by subdividing the image in blocks of a pre-set size

- Try implementing full histogram equalization, local histogram equalization, and local adaptive histogram equalization
- Local *adaptive*: use a moving window to recompute the histogram equalization



(a)



(b)



(c)

Figure 3.8 *Locally adaptive histogram equalization: (a) original image; (b) block histogram equalization; (c) full locally adaptive equalization.*

Linear filtering



45	60	98	127	132	133	137	133
46	65	98	123	126	128	131	133
47	65	96	115	119	123	135	137
47	63	91	107	113	122	138	134
50	59	80	97	110	123	133	134
49	53	68	83	97	113	128	133
50	50	58	70	84	102	116	126
50	50	52	58	69	86	101	120

$f(x,y)$

*

0.1	0.1	0.1
0.1	0.2	0.1
0.1	0.1	0.1

$h(x,y)$

=

69	95	116	125	129	132
68	92	110	120	126	132
66	86	104	114	124	132
62	78	94	108	120	129
57	69	83	98	112	124
53	60	71	85	100	114

$g(x,y)$

- In other words image is modified according to a certain kernel
- Kernel allow many operations such as: blurring (low-frequency filtering), edge detection (high-frequency filtering),

Kernel examples



$$\frac{1}{K^2}$$

1	1	...	1
1	1	...	1
⋮	⋮	1	⋮
1	1	...	1

$$\frac{1}{16}$$

1	2	1
2	4	2
1	2	1

$$\frac{1}{256}$$

1	4	6	4	1
4	16	24	16	4
6	24	36	24	6
4	16	24	16	4
1	4	6	4	1

$$\frac{1}{8}$$

-1	0	1
-2	0	2
-1	0	1

$$\frac{1}{4}$$

1	-2	1
-2	4	-2
1	-2	1

$$\frac{1}{K}$$

1	1	...	1
---	---	-----	---

$$\frac{1}{4}$$

1	2	1
---	---	---

$$\frac{1}{16}$$

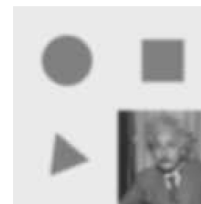
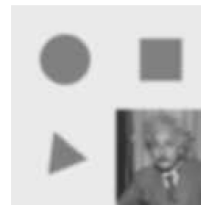
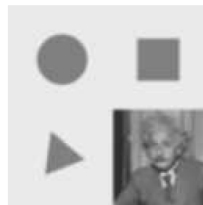
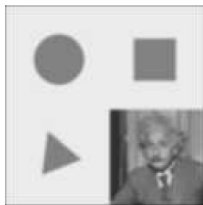
1	4	6	4	1
---	---	---	---	---

$$\frac{1}{2}$$

-1	0	1
----	---	---

$$\frac{1}{2}$$

1	-2	1
---	----	---



(a) box, $K = 5$

(b) bilinear

(c) “Gaussian”

(d) Sobel

(e) corner

Band-pass filters

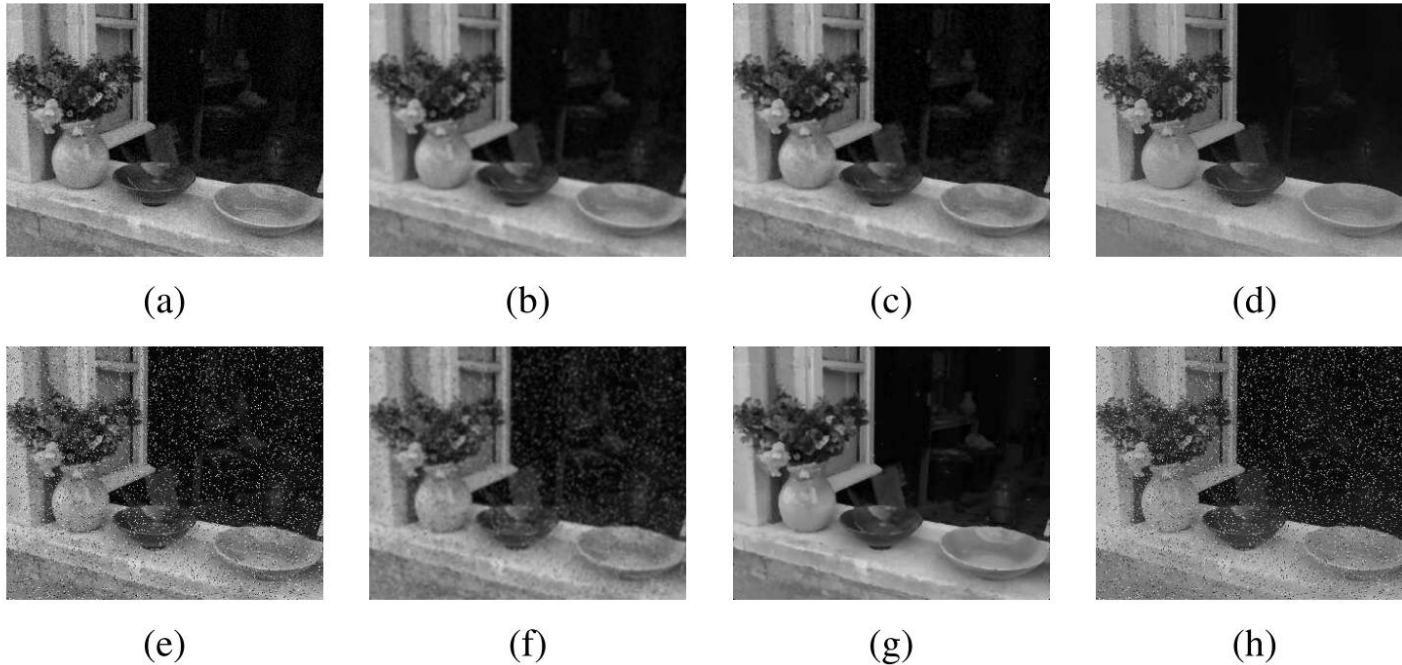
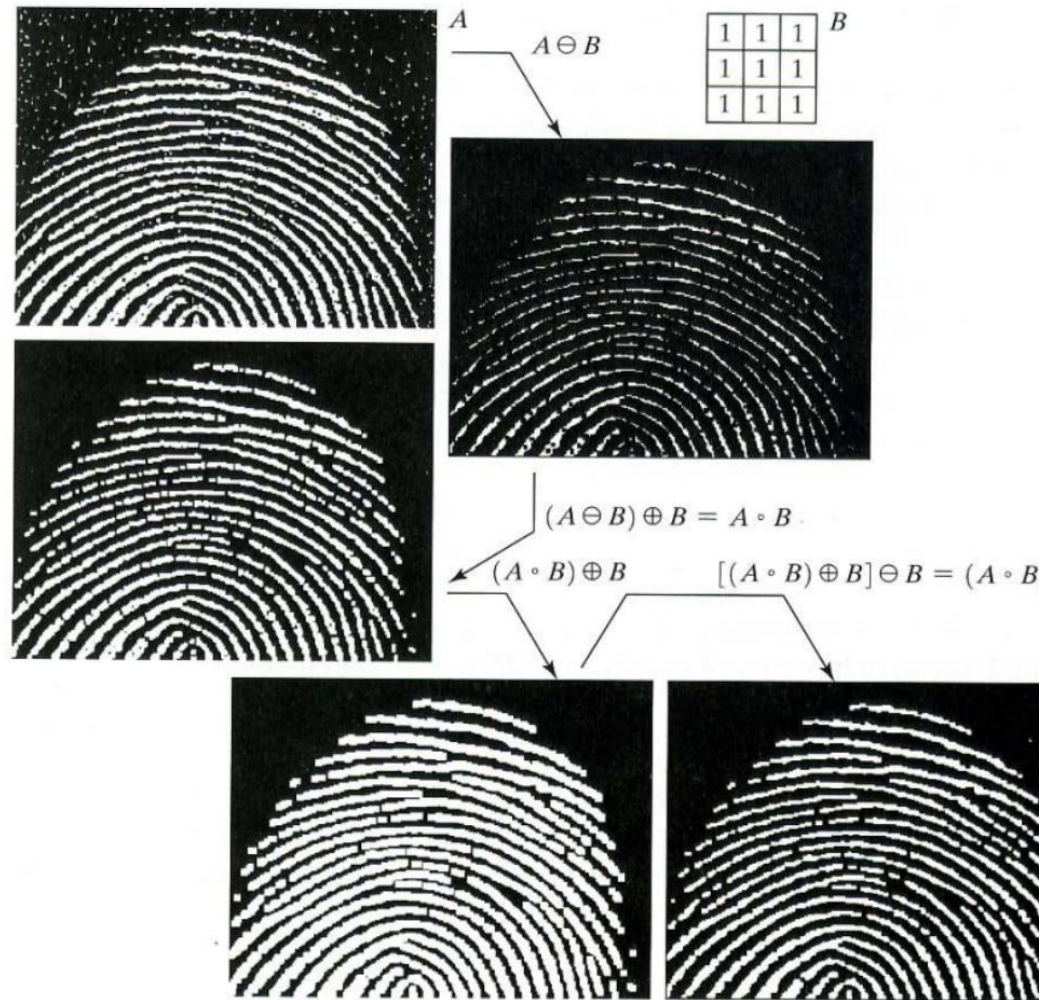


Figure 3.18 *Median and bilateral filtering: (a) original image with Gaussian noise; (b) Gaussian filtered; (c) median filtered; (d) bilaterally filtered; (e) original image with shot noise; (f) Gaussian filtered; (g) median filtered; (h) bilaterally filtered. Note that the bilateral filter fails to remove the shot noise because the noisy pixels are too different from their neighbors.*

Morphology



- Applied to binarized images
- They change the resulting shape
- Erosion
- Dilation
- Opening
- Closure



- **Tabulation of magnitude and phase for each frequency**

$$H(k_x, k_y) = \frac{1}{MN} \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} h(x, y) e^{-j2\pi(k_x x/M + k_y y/N)}$$

- **Fast Fourier Transform to reduce computational complexity from $O(N^2)$ to $O(N \log_2 N)$**

$$F(k, l) = \sum_{i=0}^{N-1} \sum_{j=0}^{N-1} \cos\left(\frac{\pi}{N}\left(i + \frac{1}{2}\right)k\right) \cos\left(\frac{\pi}{N}\left(j + \frac{1}{2}\right)l\right) f(i, j)$$

- **Applications: sharpening, blur, noise removal (and feature extraction...)**



...and finally, random variables!

Event nature and their «probability of occurrence»



- Real («tomorrow it will rain») or logical (« $X > 1$ »)
- Events can be «connected» each others
 - $A = \text{«tomorrow it will rain»}$
 - $B = \text{«tomorrow I will take my umbrella»}$
 - $A \cup B = \text{«tomorrow it will rain OR I will take my umbrella»}$
 - $A \cap B = \text{«tomorrow it will rain AND I will take my umbrella»}$
- We may associate to each event a probability of occurrence, which is a number in the interval $[0,1]$
 - $P(A) = \text{probability that tomorrow it will rain}$
 - $P(B) = \text{probability that I will take my umbrella}$
- «Certain event»: often identified with the symbol Ω , such that $P(\Omega)=1$

Probability computation

- **Classical**

- When we dispose of all possible cases of occurrence («certain event», «favourable cases»)
 - Occurences are mutually exclusive
- $$P(A) = \frac{\#FavourableOccurrences}{\#Occurrences}$$

- **Frequency**

- Based on the «history» of the event along time
- We do not know how many are the favourable cases

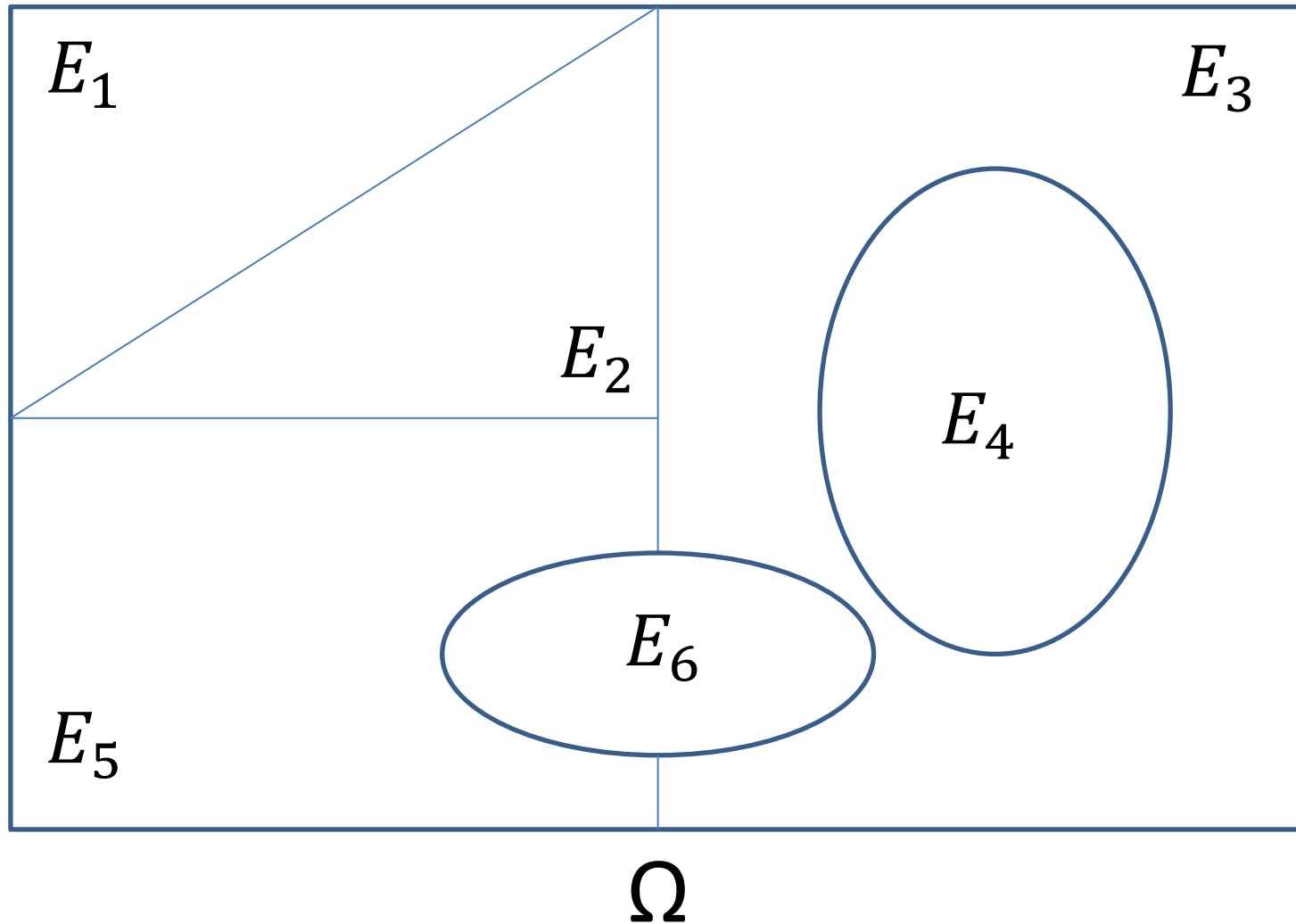
$$P(A(t)) = \frac{\#FavourableOccurrences(t)}{\#Occurrences(t)}$$

- **Subjective**

- We associate the value according to internal or external measurements associated to the event or even our «feelings»

$$P(A(x)) = \frac{\#FavourableOccurrences(x)}{\#Occurrences(x)}$$

Event space and their relationships



Main laws



- Given two events A, B :

- $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

- If $A \cap B = \emptyset$, $P(A \cap B) = 0$

- If $\{E_1, \dots, E_N\}$ is a partition of A , such that

$$\forall i, j: E_i \cap E_j = \emptyset,$$

$$\bigcup_{i=1}^N E_i = A$$

- Then:

$$\sum_{i=1}^N P(E_i) = P(A)$$

$P(A \cap B)$: Joint probabilities



- When we want to assess the probability of intersection between two events, we say that we want to assess their joint probability of occurrence

$$P(A \cap B) = P(A \cdot B)$$

- An important concept is the one about two events may be called as «independent», where:

$$P(A \cdot B) = P(A) \cdot P(B)$$

- **Logical** product between two events $\rightarrow$ **arithmetical** product between their individual probabilities

$P(A|B)$: conditional probability



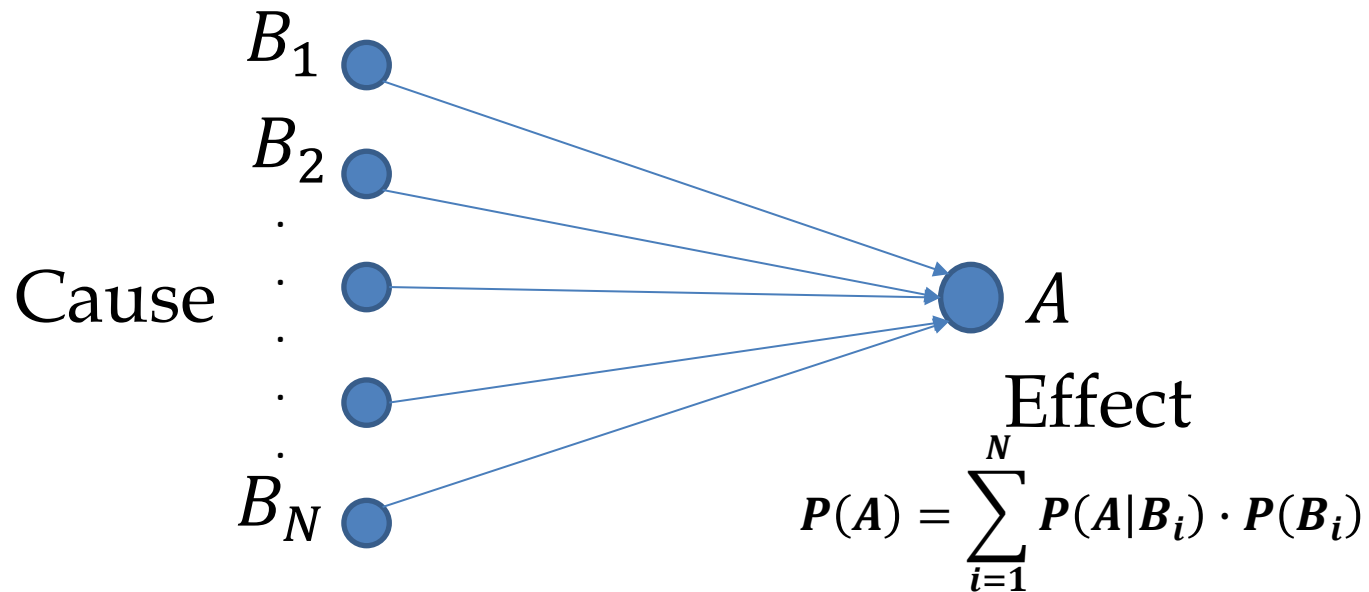
$$P(A|B) = \frac{P(A \cdot B)}{P(B)}$$

- This leads to important consequences as:

$$P(A) = \sum_{i=1}^N P(A|B_i) \cdot P(B_i) = \sum_{i=1}^N P(A \cdot B_i)$$

- If $\{B_1, \dots, B_N\}$ is a partition of a generic event B

$$P(A|B_i) = \frac{P(B_i|A) \cdot P(A)}{P(B_i)}$$





- **Numerical representation of an event**
 - E.g. coin toss $\rightarrow$ head/cross $\rightarrow x=\{1,0\}$
- **R.V. can be discrete or continuous**
- **As it is possible to associate a probability function to an event, the same hold for random variables, namely, $x \rightarrow$ probability distribution $p(x)$**
- **The probability distribution maps the value of a r.v. to the $[0,1]$ interval**
- **The sum of $p(x)$ for any value of x is equal to 1**

- **Bernoulli**

- Given a boolean r.v. $x \in \{0,1\}$
- $P(X = x; p) = p^x(1 - p)^{1-x}$

- **Fraud Detection**

- Suppose a bank wants to determine whether a specific transaction is fraudulent (1) or legitimate (0).

- **Biometric Authentication**

- When a fingerprint scanner verifies a user, the system either accepts (1) or rejects (0) the authentication attempt.



- **Binomial**
 - Given a r.v. $x \in \{0, 1, \dots, N\}$
 - $P(X = x; p) = \binom{N}{x} p^x (1 - p)^{N-x}$
- **Deepfake Detection Accuracy**
 - Suppose a forensic AI system evaluates 100 videos and each video has a 20% probability of being a deepfake. The binomial distribution can model the number of deepfake videos detected.
- **Fingerprint Authentication Attempts**
 - If a fingerprint scanner is used by 50 people per day, and the probability of a false rejection (failure) is 5%, the binomial distribution models how many users might be falsely rejected in a day.

- **Poisson**

- Given a r.v. $x \in \mathbb{N}$
- $P(X = x; \lambda) = \frac{\lambda^x}{x!} e^{-\lambda}$

- **Identity Fraud Cases per Month**

- If on average, 5 cases of identity fraud are detected per month in a country, a Poisson model can estimate the probability of detecting, say, 7 or 10 cases in a given month.

- **Biometric System Errors per Hour**

- If a biometric entry system in an airport processes 5000 people per day and experiences 3 recognition failures per hour, the Poisson distribution can estimate how often errors occur.

- **Uniform**

- Given a r.v. $x \in [a, b]$

- $p(x; a, b) = \frac{1}{b-a} I(x)$, where $I(x) = \begin{cases} 1, & x \in [a, b] \\ 0, & x \notin [a, b] \end{cases}$

- **Random Key Generation in Cryptography**

- If an encryption system generates a **random key** between 1 and 1000, a **uniform distribution** ensures that each key has an **equal chance** of being selected.

- **Face Positioning in Biometric Image Capture**

- When a facial recognition system captures images, the position of a person's face on the screen (within a certain bounding box) may be **uniformly distributed** if users are randomly positioned.

- **Random Sampling for Forensic Analysis**

- When selecting **N random images** from a dataset for forensic examination, a **uniform distribution** ensures that each image has an **equal probability** of being chosen.

- **Gaussian**

- Given r.v. $x \in \mathfrak{R}$
- $p(x; \mu, \vartheta) = \frac{1}{\sqrt{2\pi\vartheta}} \exp\left(-\frac{(x-\mu)^2}{2\vartheta^2}\right)$

- **Fingerprint Ridge Thickness Variability**

- The thickness of fingerprint ridges in a population follows a normal distribution, meaning most people have average ridge thickness, while very thick or very thin ridges are rare.

- **Face Recognition Similarity Scores**

- Biometric face-matching systems produce similarity scores that often follow a Gaussian distribution, where most genuine matches have scores close to the mean, while few are very high or very low.

- **Reaction Time in Security Systems**

- In an AI-based anomaly detection system, the response time to process an alert may be normally distributed, with most responses clustering around an average processing time.

- **Beta**

- Given r.v. $x \in [0,1]$
- $p(x; \alpha, \beta) = \frac{x^{\alpha-1}(1-x)^{\beta-1}}{B(\alpha, \beta)}$, where $B(\alpha, \beta) = \int_0^1 x^{\alpha-1}(1-x)^{\beta-1} dx$

- **Deepfake Detection Confidence**

- If an AI model has analyzed 1000 videos and found that 30% are deepfakes, a Beta distribution helps estimate the true proportion of deepfakes in a larger dataset with uncertainty.

- **False Rejection Rate (FRR) in Biometrics**

- If a biometric authentication system has been tested on 200 users, and 10 were falsely rejected, the Beta distribution helps predict the probability of future false rejections.

- **AI Trustworthiness in Decision-Making**

- Suppose an AI forensic tool has classified 500 out of 2000 images as manipulated, but we're uncertain about its accuracy. A Beta distribution can help model the probability that a new image is actually manipulated.

- **Multinomial**

- Given $(x_1, \dots, x_M) \in \{0, 1, \dots, N\}^M$, $\sum_{i=1}^M x_i = N$
- $P((X_1 = x_1, \dots, X_M = x_M); (p_1, \dots, p_M)) = \frac{N!}{\prod_{i=1}^M x_i!} \prod_{i=1}^M p_i^{x_i}$

- **Face Attribute Classification in Biometrics**

- Suppose an AI model classifies faces into categories like "Male," "Female," "Child," "Elderly," and "Uncertain" based on facial features. The number of detected faces in each category follows a Multinomial distribution

- **Deepfake Type Detection**

- If a forensic tool analyzes 100 videos, categorizing each as "Real," "AI-Synthesized," "Face-Swapped," or "Re-Touched", the number of cases in each category follows a Multinomial distribution

- **Document Type Classification in ID Verification**

- A biometric-based document verification system classifies incoming ID scans as "Passport," "Driver's License," "National ID," or "Other." The frequency of each type in a batch of 1000 scans follows a Multinomial distribution.

- **Multivariate gaussian**
 - Given $(x_1, \dots, x_M) \in \mathbb{R}^M$
 - $P(\mathbf{x} = (x_1, \dots, x_M); \boldsymbol{\mu} = (\mu_1, \dots, \mu_M), \boldsymbol{\Sigma}) = \frac{\exp[(\mathbf{x} - \boldsymbol{\mu})^T \boldsymbol{\Sigma}^{-1} (\mathbf{x} - \boldsymbol{\mu})]}{\sqrt{(2\pi)^M |\boldsymbol{\Sigma}|}}$
- **Facial Feature Analysis for Identity Verification**
 - In a biometric recognition system, face embeddings (e.g., eye distance, nose width, mouth shape) follow a Multivariate Gaussian distribution, since facial features are correlated
- **Fingerprint Pattern Analysis**
 - If a system analyzes ridge curvature, minutiae density, and ridge orientation, these measurements are often normally distributed and correlated, making a Multivariate Gaussian model useful
- **Voice Biometrics for Speaker Recognition**
 - Features like pitch, vocal intensity, and speech rate in speaker verification tend to follow a Multivariate Gaussian distribution, as they vary together in predictable ways
- **AI-Based Anomaly Detection in ID Fraud**
 - If an AI system evaluates multiple biometric and non-biometric features (e.g., face match score, fingerprint quality, typing speed), the combined feature set can be modeled as a Multivariate Gaussian distribution to detect anomalies.



- Let's consider $P(A)=0.35$, $P(B)=0.45$, $P(A+B)=0.6$. Verify if A and B can be claimed as independent. If not, compute $P(A | B)$ and $P(B | A)$.
- A biometric fingerprint scanner has a 95% accuracy in recognizing a legitimate user. If a user attempts to unlock their phone, define the Bernoulli random variable and calculate (1) the probability that the user is correctly recognized and (2) the probability that the system rejects them even though they are legitimate.
- An identity fraud detection system detects an average of 4 weekly fraudulent cases. What is the probability that in a given week, exactly 6 fraud cases are detected? What is the probability that no fraud cases are detected in a week?

- **Bernoulli**

- $P(\text{correct recognition}) = 0.95$
- $P(\text{rejection}) = 1 - 0.95 = 0.05$
- Since this is a **Bernoulli trial**, there is only **one attempt** with two possible outcomes. The probability is directly given and does not require further calculations.

- **Poisson**

- Develop the Poisson expression such that $\lambda=4$ (expected fraud cases per week), k is the number of fraud cases we want to find.
- Note: Poisson models the number of **random, rare events** occurring **within a fixed time period**.

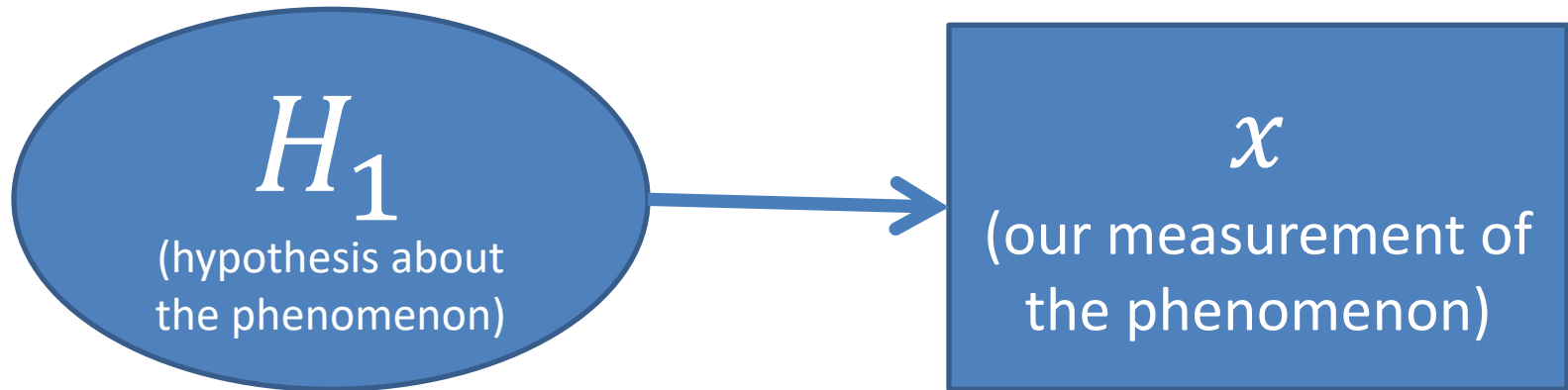
- **Values which synthesize the main properties of the distributions**
- **Mean**
 - $\mu = E(X) = \sum_i x_i P(x_i)$ or $\int x \cdot p(x) dx$
 - x_i is here one of the possible values in the r.v.'s domain
- **Variance**
 - $\sigma^2 = E\left((X - E(X))^2\right) = \sum_i (x_i - \mu)^2 P(x_i)$ or $\int (x - \mu)^2 \cdot p(x) dx$

Hypothesis verification tests



- The hypothesis verification test is aimed to confirm or deny an hypothesis on the event described by the r.v.
- Such hypothesis is called “NULL” hypothesis
- Tests can be:
 - Single: if the NULL hypothesis is not verified, then the “alternate” hypothesis is verified
 - Sequential: if the NULL hypothesis is not verified, then:
 - The “alternate” hypothesis is verified, OR
 - There is not enough evidence, and the test must be repeated
➔ SEQUENTIAL HYPOTHESIS VERIFICATION TESTS

Simple likelihood based classification/hypothesis verification test



We can simply evaluate, for each value of x :

- how many times x is measured when H_1 is true, so computing the probability $p(x|H_1)$
- how many times x is measured when H_1 is false, so $p(x|H_0)$

Maximum Likelihood criterion: Decide that H_1 is true if, according to a novel measurement x^{new} , $p(x^{new}|H_1) > p(x^{new}|H_0)$

A posteriori hypothesis verification test

- Let's suppose that we have some a priori knowledge about our hypothesis H_1 , which is represented by the probability of occurrence $P(H_1)$, so that $P(H_0)=1-P(H_1)$

- By applying the Bayes' theorem:

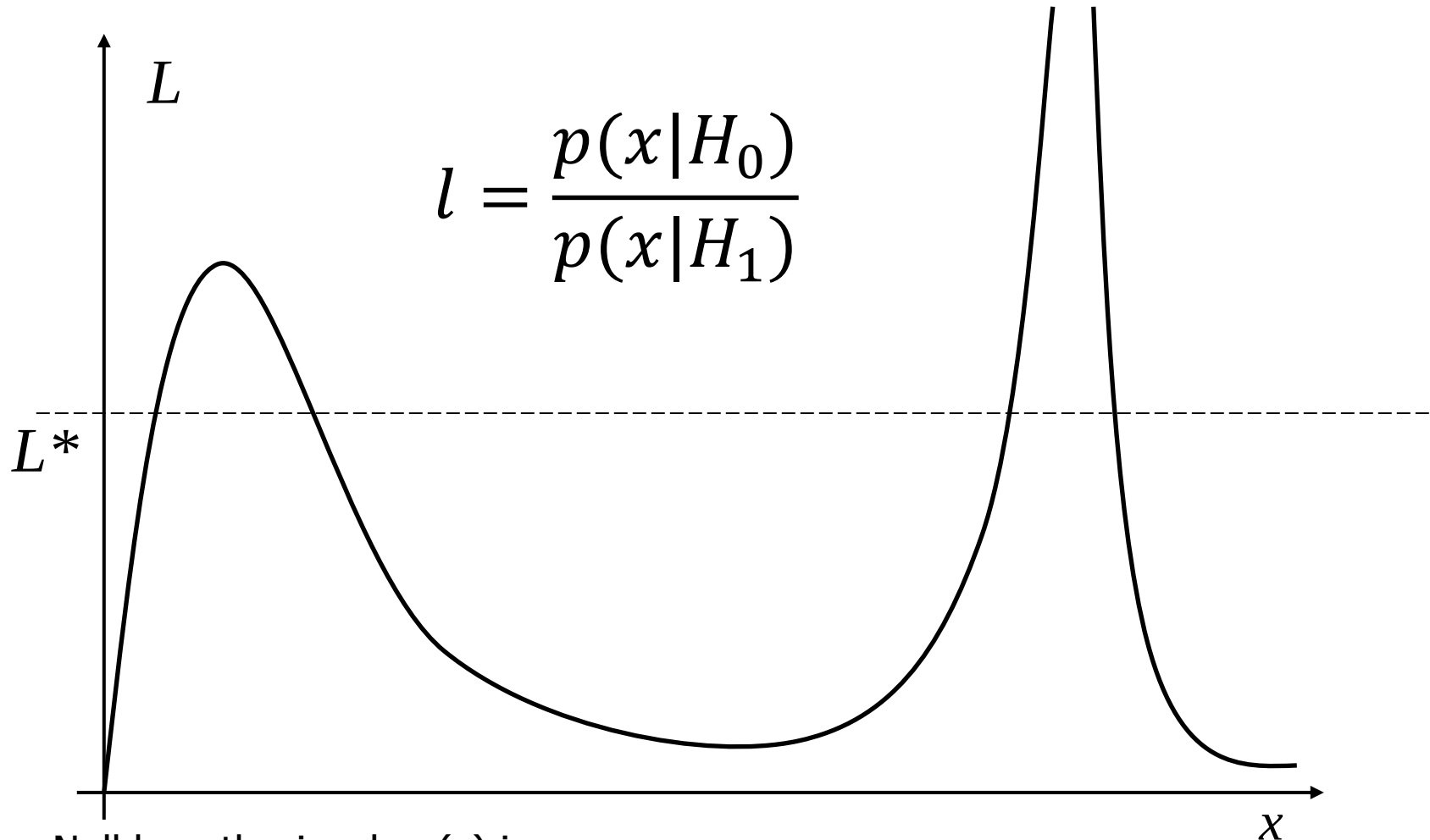
$$P(H_1|x) = \frac{p(x|H_1) \cdot P(H_1)}{p(x)}$$

- Maximum a posteriori criterion: decide for H_1 if $P(H_1|x) > P(H_0|x)$

Property of the Neyman-Pearson test

- **Errors of a hypothesis verification test:**
 - False Acceptance (FAR): the NULL hypothesis is wrongly verified
 - False Rejection (FRR): the NULL hypothesis is wrongly rejected
- **NP-test is the most powerful test in the following terms:**
 - For each test t such that its FAR is equal to that of NP-test, the correspondent FRR of NP-test is **always** lower than that of t

Neyman-Person test

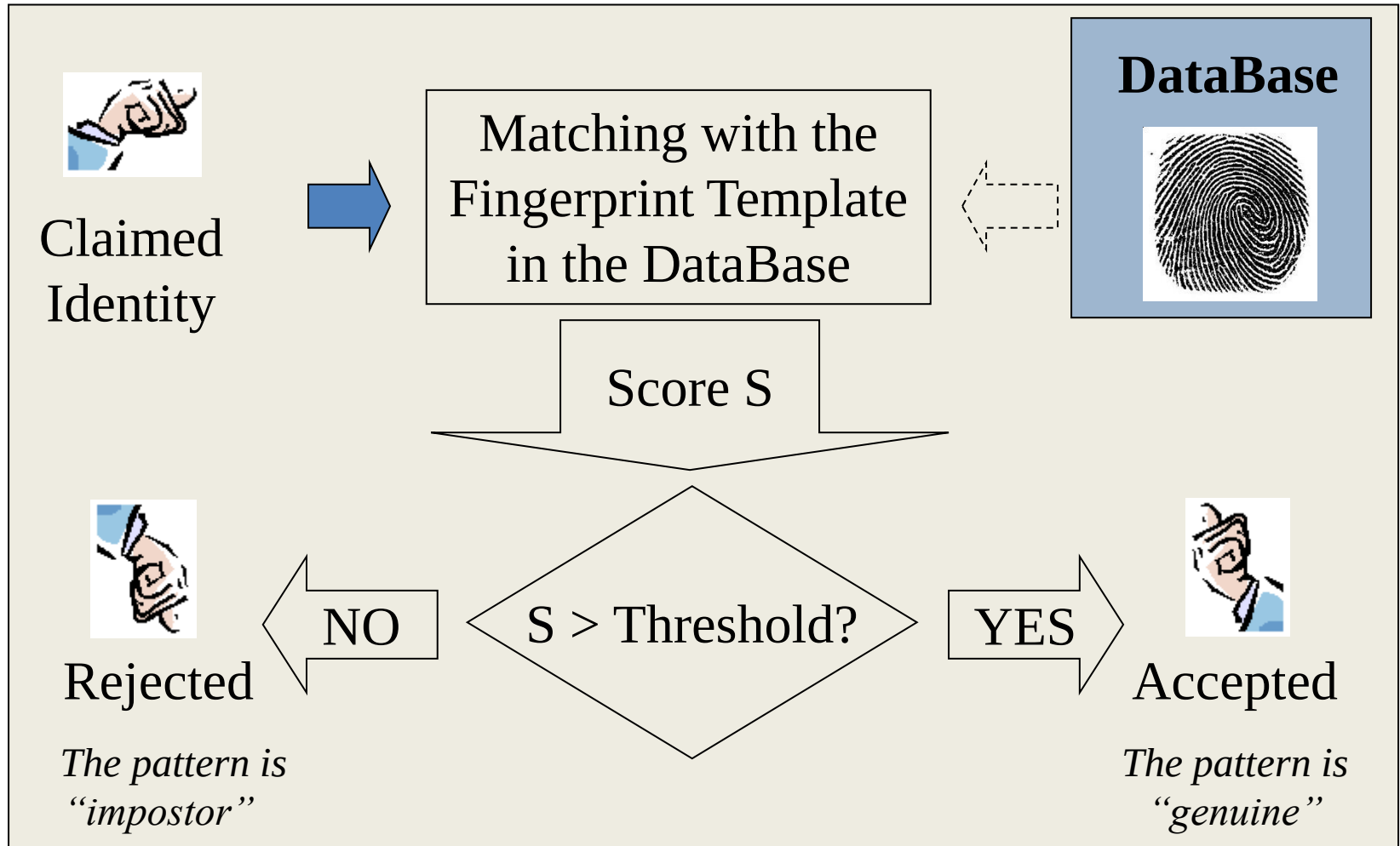


Null hypothesis: class(x) is ω_0

Alternate hypothesis : class(x) is ω_1

Null hypothesis is verified if $l > L^*$

An easier test for personal authentication



That's all for today... what's next?

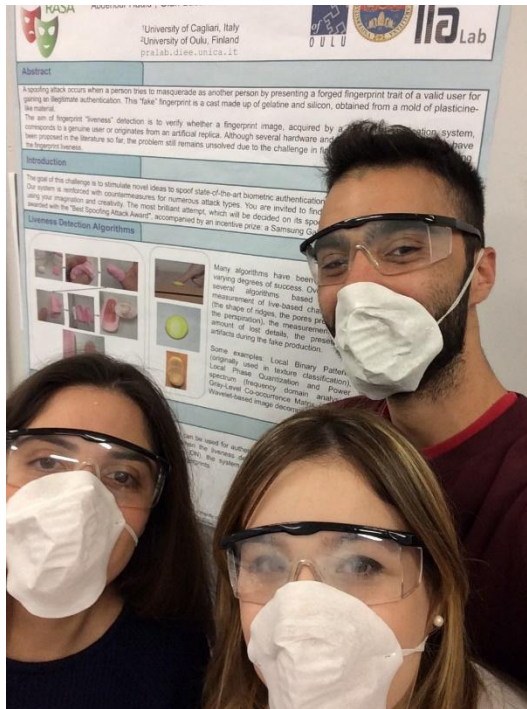


- **Introduction to biometric technologies**
- **Fingerprints**
 - What are they?
 - Main tasks
- **Fingerprint recognition system**
 - Pre-processing
 - Feature extraction
 - Matching
- **Fingerprint liveness detection**
 - The problem
 - State of the art
 - Solutions





Thank you for listening!



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